# LITERATURE SURVEY REPORT

# Title: Student Feedback Analysis Using Natural Language Processing (NLP) and Sentiment Analysis

## 1. Introduction

This paper focuses on the importance of student feedback in improving the quality of education in academic institutions. Typically, course evaluations include both quantitative ratings and qualitative comments. While numerical ratings are easy to analyze and provide quick insights, qualitative feedback contains deeper information about student experiences, such as teaching effectiveness, engagement, and clarity of instruction.

However, analyzing these written comments manually is time-consuming and complex, which often leads to them being underutilized. To address this challenge, the study introduces sentiment analysis, a technique from Natural Language Processing (NLP), which can automatically classify feedback into positive, negative, or neutral sentiments. By applying this approach, the research aims to make qualitative feedback more structured and useful, enabling institutions to better understand student perceptions and improve teaching and learning outcomes.

## 2. Objective of the Study

The main goals of this research are:

- To classify student feedback into **positive, negative, and neutral sentiments**
- To analyze the relationship between **student comments and their ratings**
- To improve understanding of **teaching effectiveness and course quality**

## 3.Methodology

The research follows a structured approach consisting of multiple stages:

### 3.1 Data Collection

- Data collected from **13 Civil Engineering courses**
- Total of **377 student comments**
- Students answered 3 key questions related to course experience

### 3.2 Data Annotation

Two types of labeling were used:

- **Manual Annotation** (human labeling)
- **Machine Annotation** using DistilRoBERTa

Differences between both were resolved through **re-annotation** for better accuracy.

.

### 3.3 Data Augmentation

Problem: Very few negative comments (data imbalance)

Solution: **Back Translation**

- Translate English → Japanese → English
- Generates more negative samples without changing meaning

### 3.4 Model Used

- **DistilRoBERTa (a lightweight BERT model)**
- Fine-tuned for sentiment classification

**Key Features:**

- Understands context of words
- Faster and efficient compared to traditional models
- Works well with smaller datasets

### 3.5 Model Training

- Dataset split:
  - **70% Training**
  - **15% Validation**
  - **15% Testing**
- Best performance:
  - Learning rate: **5e-6**
  - Epochs: **30**
  - Accuracy: **90%**

### 3.6 Statistical Analysis

Two main techniques were used:

**1. Pearson Correlation**

- Result: **0.42**
- Indicates a **moderate positive relationship**
- Meaning: More positive comments → Higher ratings

**2. Linear Regression**

- Coefficient: **0.24**
- Meaning: Increase in sentiment leads to slight increase in ratings

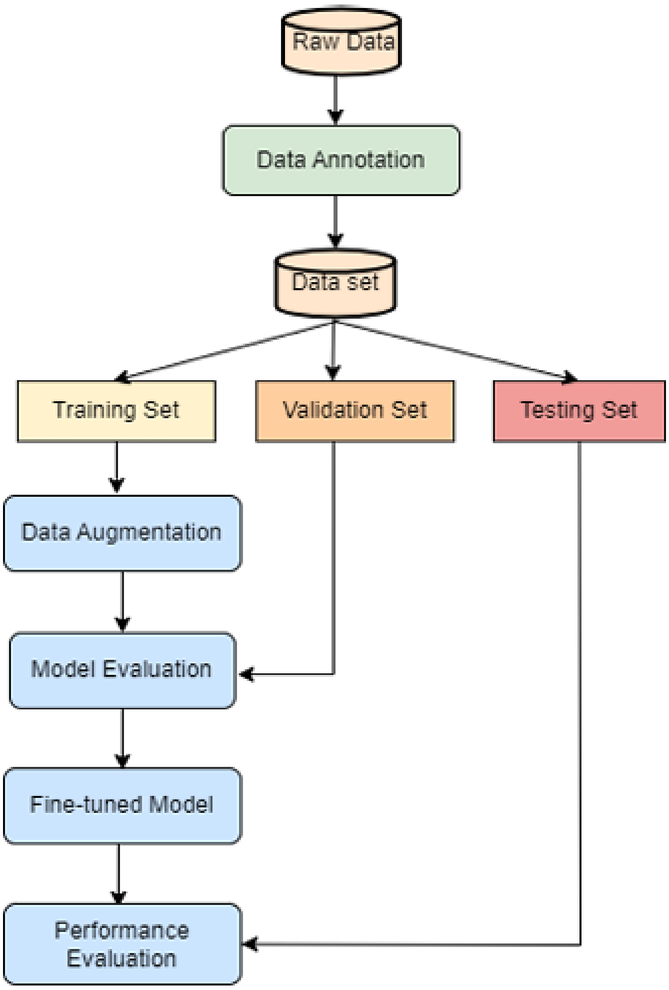


**Figure 1 Proposed methodology for sentiment analysis**

## 4. Results

- Positive comments: **147**
- Neutral comments: **152**
- Negative comments (after augmentation): **132**

The chart on **page 10** shows how data became more balanced after augmentation.

- Model achieved **90% accuracy**
- Sentiment analysis successfully identified patterns in feedback

## 5. Discussion

- Sentiment and ratings are **related but not strongly dependent**
- Other factors like:

- Teaching style
- Course difficulty
- Studentexpectations
  also influence ratings

## 6. Conclusion

- Sentiment analysis is an effective tool for analyzing student feedback
- Helps convert text into measurable insights
- However, it cannot fully replace traditional evaluation methods
- Best results come from combining **qualitative + quantitative data**

## 7. Practical Implications

**For Institutions:**

- Helps identify poorly rated courses early
- Supports data-driven decision making

**For Faculty:**

- Provides quick overview of student opinions
- Helps improve teaching methods

## 8. Challenges

- Small dataset size
- Imbalanced data (few negative comments)
- Subjectivity in labeling feedback
- Time-consuming annotation process

## 9. Future Scope

- Use larger and more diverse datasets
- Apply **aspect-based sentiment analysis**
- Include audio/video feedback
- Develop real-time feedback systems

This research uses NLP and a deep learning model called DistilRoBERTa to analyze student feedback. It classifies comments into sentiments and compares them with student ratings. The model achieved 90% accuracy and showed a moderate correlation between feedback and ratings. The study proves that sentiment analysis can help improve course evaluation, but it should be used along with other methods for better results.